INFO 523 Data Mining and Discovery Final Project REPORT

Analysis of Emotions Using Keystroke Dynamics and Typing Behaviors

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1. Introduction:

This research investigates a unique keystroke dynamics dataset, depicting natural emotional experiences in the realm of human-computer interactions, and psychological research. Therefore, this research uses data taken from a comprehensive EmoS (Emotion Survey) dataset that details much information about participants, considering their emotional states and demographics, and their characteristics for typing behavior. The dataset is rich in variables such as emotions:

- Neutral.
- Happy.
- Sad.
- Angry.
- Calm.

Also considers variables of age ranges; response types; and key press frequencies that provide a firm ground for investigating the intricate relationships between emotional states and cognitive performance.

The current research effort has taken up a robust data mining approach for answering three of the most important research questions: firstly, exploring the relationship between the emotional states and response accuracy; secondly, analysis of emotional experiences across different age groups; and thirdly, the relationship between typing speed and emotional indices. By using high-level statistical methods and visualizations, this study endeavours to unpack the subtle ways in which emotions can impact cognitive processing, typing behaviors, and decision-making. The results contribute not only to understanding human-computer interaction better but also provide a possibly profound insight into emotion recognition, cognitive performance, and complex interplays between emotional states and behavioral patterns.

1.1 About the DATASET:
The dataset contains 4 .csv files:

- *File 1:* Fixed Text Typing Dataset which is collected while a participants it typing a fixed text and it includes the following features: User Id, Emotion Index, Index, Key Code, key Down, key Up, D1U1, D1U2, D1D2, U1D2, U1U2, D1U3, D1D3, and Answer.
- *File 2:* Free Text Typing Dataset which is collected while a participants it typing a free text and it includes the following features: User Id, Emotion Index, Index, Key Code, key Down, key Up, D1U1, D1U2, D1D2, U1D2, U1U2, D1U3, D1D3, and Answer.
- *File 3:* Frequency Dataset which includes frequency related features like User ID, textIndex, EmotionIndex, DelFreq, LeftFreq, and TotTime.
- *File 4:* Participants Information dataset which includes demographics information like UserID, TypeWith, TypistType, PCTimeAverage, AgeRange, gender, status, degree, and country.

Data extracted from: Aicha Maalej, Ilhem Kallel. (2020). EmoSurv: A typing biometric (Keystroke dynamics) dataset with emotion labels created using computer keyboards. IEEE Dataport. https://dx.doi.org/10.21227/eae6-pk42

2. Question 1: What emotional characteristics are predominantly exhibited by participants who provide correct responses?

2.1 Introduction:

Our first question involves the complex relationship between emotional states and cognitive performance which has long been a source of interest in the fields of cognitive psychology and behavioral research. We try to focus the emotional state of participants alongside their response accuracy, thus offering a rich opportunity to explore how different emotional contexts might influence decision-making and response quality. By focusing on the key columns of "emotionIndex" and "answer" types, we want to explore potential links that may exist between emotional experience and cognitive performance.

Our interest in this research question is based on its potential to reveal how emotional states might modulate cognitive processes and accuracy. By looking at which emotional characteristics are most closely associated with the correct responses, we might get an idea of how different affective states might influence a person's ability to answer questions correctly.

1.2 Approach/Methodology:

1.2.1 ***Preprocessing and Cleaning****:*

-The first step was uploading the CSV and setting up a Pandas data frame to work in.
- Here we are using Free Text Typing Dataset.csv.

-Renaming the columns.

-Drop any rows with missing "emotionIndex" or "answer" values. (This step is very important in ensuring the analysis will be done on complete and valid data points).

1.2.2 ***Filtering and Selecting Relevant Data****:*

-Filter the DataFrame for those rows where the "answer" column is equal to "R" (Right Answer).

2.2.3 ***Counting Frequencies and Occurrence Analysis****:*

-Create the frequency count of every emotion index for those rows where the answer is correct.

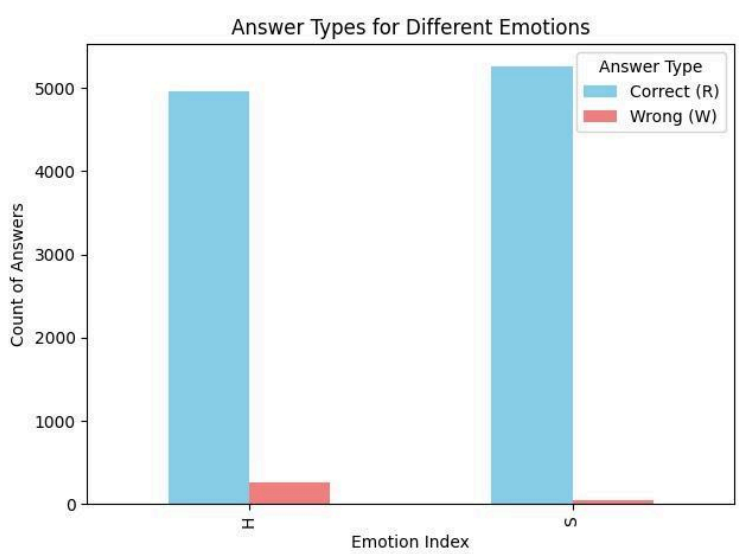
-Group data by "emotionIndex" and "answer" and count the number of occurrences of every pair of values.

2.2.4 ***Data Visualization****:*

-Uses Matplotlib.pyplot to create bar charts that visualize "Answer Types for Different Emotions" and "Representation of Answer Types by Emotion".

1.3 Data Analysis and Results:

```
# Plot the results as a bar chart
emotion_answer_counts.plot(kind='bar', stacked=False, color=['skyblue', 'lightcoral'])
plt.title("Answer Types for Different Emotions")
plt.xlabel("Emotion Index")
plt.ylabel("Count of Answers")
plt.legend(title="Answer Type", labels=["Correct (R)", "Wrong (W)"])
plt.tight_layout()
plt.show()
```



Plot 1. Answer Types for Different Emotions.

1.4 Discussion:

From plot 1 we can check that the predominant emotions in the datasets are Happiness and Sadness, which:

1. The correct answer for happy emotions is way larger than the wrong ones. It seems like, in general, when it comes to high-intensity emotions, participants tended to give a correct response.

2. The number of correct answers also outnumbers the wrong answers for sadness, but to a lesser extent compared to happy emotions. This means participants found it relatively easier to give the correct response for low-intensity or subtle emotions.

Keeping this in mind, the data suggest that people may find it easier to identify and respond appropriately to low-intensity emotions than to high-intensity emotions. This may be due to a variety of factors, including the salience of emotional cues, the level of emotional awareness and sensitivity, or the cognitive processes involved in emotion recognition and response.

2. Question 2: How do emotional states (emotionIndex) vary across different age ranges?

3.1 Introduction:

Another important thing to bring to the table is understanding the interplay between the emotional states of an individual and their age is very important in psychological and behavioral research. With this particular dataset, we got the chance to investigate this relationship since it captures detailed information about the participants' emotional states, as well as their demographic characteristics, including age.

By examining how emotional states, as represented by the "emotionIndex" variable, vary across different age ranges, we can gain valuable insights into the dynamic nature of human emotions and how they evolve over the course of an individual's lifespan. This study is in line with the general objective of the research project, which is to investigate the feasibility of using keystroke dynamics as a new approach to emotion recognition and classification.

2.2 Approach/Methodology:

3.2.1 ***Data Preprocessing and Merging****:* Merge the two separate datasets, "Participants Information.csv" and "Frequency Dataset.csv" based on the "User_ID" column.

3.2.2 ***Handling Missing Data****:* We address the issue of missing data by dropping rows with NaN (Not a Number) values in the "emotionIndex" column.

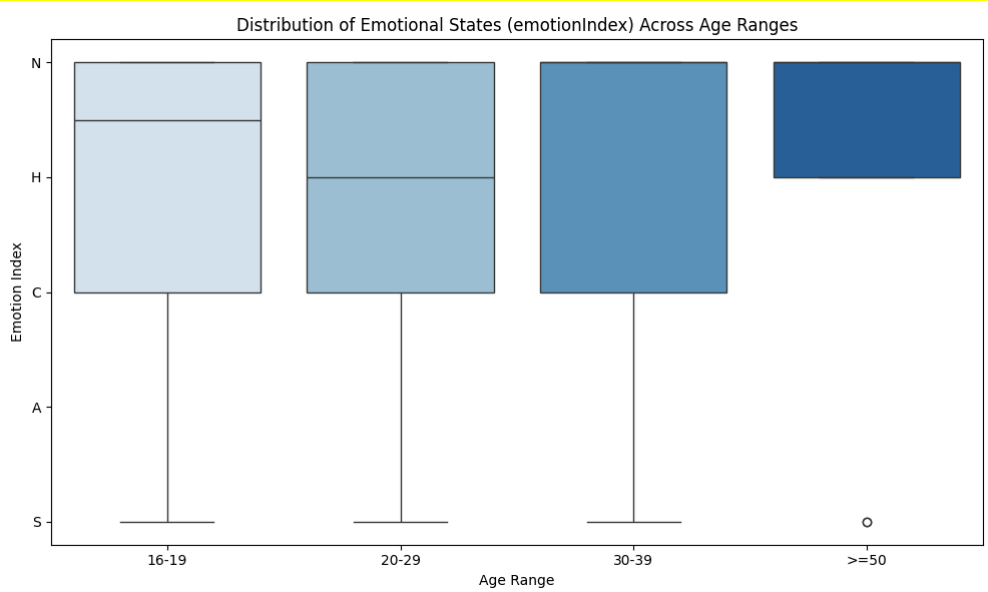
3.2.3 ***Calculating Average Emotional State by Age Range****:* Calculate the average emotional state, represented by the "emotionIndex" variable, for each age range.

3.2.4 ***Exploring Unique Values and Missing Data****:* Examine the unique values present in the "emotionIndex" and "ageRange" columns, as well as check for any missing values in these variables.

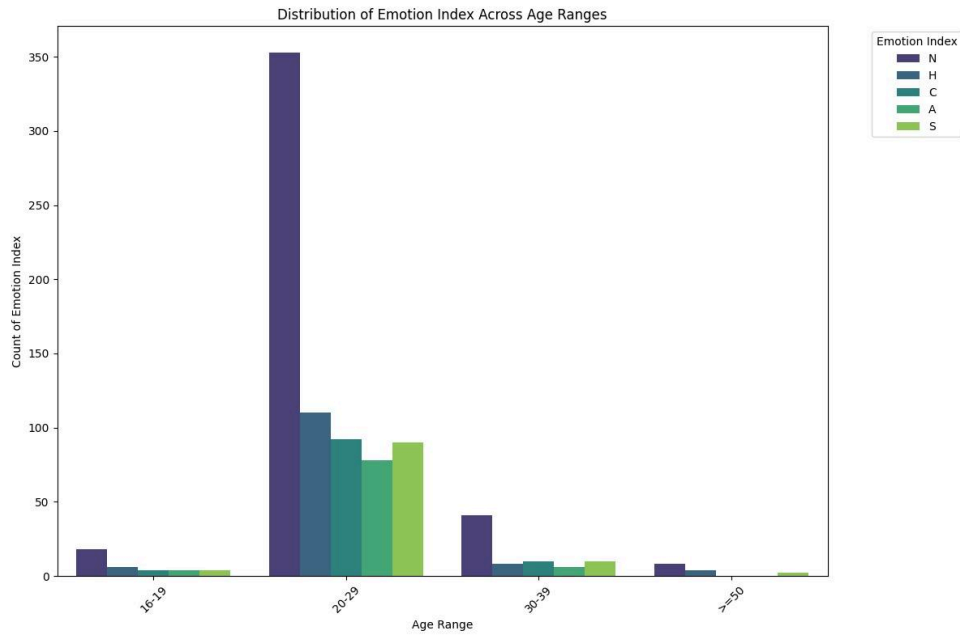
3.2.5 ***Visualizing the Distribution of Emotional States across Age Ranges****:* Using the Seaborn library, we create a bar chart that visualizes the distribution of emotional states, as represented by the "emotionIndex" variable, across different age ranges.

2.3 Data Analysis and Results:

```
# Boxplot for emotionIndex across different ageRanges
plt.figure(figsize=(10, 6))
sns.boxplot(x='ageRange', y='emotionIndex', data=merged_df, palette='Blues')

# Add labels and title
plt.title('Distribution of Emotional States (emotionIndex) Across Age Ranges')
plt.xlabel('Age Range')
plt.ylabel('Emotion Index')

# Show the plot
plt.tight_layout()
plt.show()
```



```python
import matplotlib.pyplot as plt
import seaborn as sns

# Plotting the count plot for emotionIndex across ageRange
plt.figure(figsize=(12, 8))
sns.countplot(x='ageRange', hue='emotionIndex', data=merged_df, palette='viridis')
plt.title('Distribution of Emotion Index Across Age Ranges')
plt.xlabel('Age Range')
plt.ylabel('Count of Emotion Index')
plt.xticks(rotation=45)
plt.legend(title='Emotion Index', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()
```



Plot 2a and 2b. Distribution of Emotions vs. Age Ranges.

2.4 Discussion:

From the data in Plot 2a and 2b we can see that in the 16-19 age range, the emotion index has a wide distribution ranging from S (Sad) to N (Neutral) and the median emotion index is around H (Happy). This is very similar to the distribution for the 20-29 age range but slightly shifted, with the median near H. Now, for the 30-39 age range the data is spread widely across the emotion index, spanning S to N, with a median near H. And ≥50 age range shows a smaller range of emotion index values, concentrated around H and N and a single outlier exists at the S level.

There are some interesting observations when the data is examined closely. The middle age ranges have a slightly higher overall "emotionIndex" compared to younger and older age groups, which may indicate that at some point in development or life stage, there is a

difference in emotional expression and regulation. For instance, middle-aged participants may have better-developed coping mechanisms or emotional maturity that would raise and even out their emotional state.

3. **Question 3: What is the relationship between typing speed (key press frequency) and emotion index?**

4.1 Introduction:

A good understanding of the inter-relationship of the subject's typing behavior and their emotional state is imperative for the full comprehension of human-computer interaction, hence the use of Keystroke Dynamics.

By investigating this further, a possible linkage between the emotion index with typing speed represented by key-pressing frequency may just reveal how one's affective experiences materialize via typing behavior where DelFreq is the Relative frequency of delete key NA, and Leftfreq is the Relative frequency of backspace key NA.

3.2 Approach/Methodology:

4.2.1 ***Data Preparation****:*
Filter and clean the data, ensuring there are no missing or erroneous values in DelFreq, LeftFreq, and emotionIndex.
Convert emotionIndex to a numeric scale (if necessary) so that it can be used for correlation and visualization.
We remove any rows where DelFreq, LeftFreq, or emotionIndex are missing to ensure clean analysis.

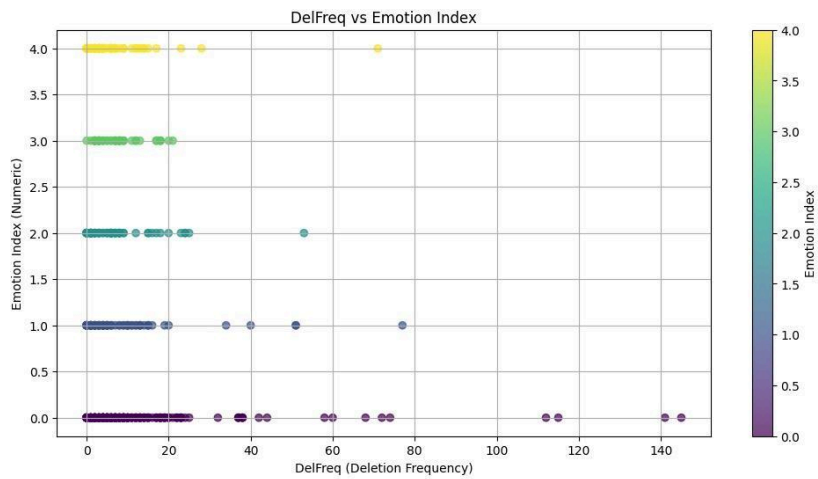
4.2.2 ***Correlation Calculation****:*
Calculate the correlation between DelFreq, LeftFreq, and emotionIndex.
The correlation between DelFreq, LeftFreq, and emotionIndex is computed using *corr()*.
The correlation values will tell us if there's any linear relationship.

4.2.3 ***Visualization****:*
We create two scatter plots:
- DelFreq vs emotionIndex
- LeftFreq vs emotionIndex

These plots will help visualize the relationships. If there's a strong positive or negative trend, we might observe a pattern in how key press frequencies relate to emotional states.

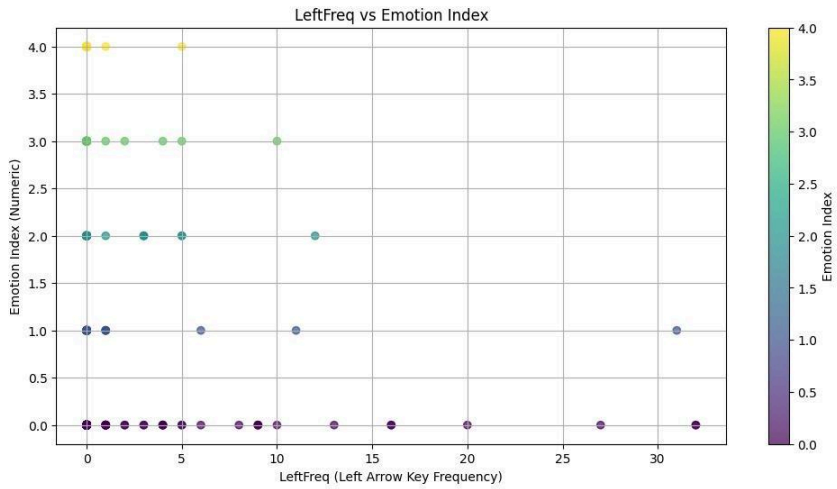
3.3 Data Analysis and Results:

```
import matplotlib.pyplot as plt

# Scatter plot for DelFreq vs emotionIndex
plt.figure(figsize=(12, 6))
plt.scatter(filtered_frequency_df['delFreq'], filtered_frequency_df['emotionIndex_numeric'],
            c=filtered_frequency_df['emotionIndex_numeric'], cmap='viridis', alpha=0.7)
plt.title('DelFreq vs Emotion Index')
plt.xlabel('DelFreq (Deletion Frequency)')
plt.ylabel('Emotion Index (Numeric)')
plt.colorbar(label='Emotion Index')
plt.grid(True)
plt.show()
```



Plot 3. DelFreq vs. Emotions.

```
# Scatter plot for LeftFreq vs emotionIndex
plt.figure(figsize=(12, 6))
plt.scatter(filtered_frequency_df['leftFreq'], filtered_frequency_df['emotionIndex_numeric'],
            c=filtered_frequency_df['emotionIndex_numeric'], cmap='viridis', alpha=0.7)
plt.title('LeftFreq vs Emotion Index')
plt.xlabel('LeftFreq (Left Arrow Key Frequency)')
plt.ylabel('Emotion Index (Numeric)')
plt.colorbar(label='Emotion Index')
plt.grid(True)
plt.show()
```



Plot 4. LeftFreq vs. Emotions.

3.4 Discussion:

From the Plots it's important to note that:

- X-axis: DelFreq (Deletion Frequency): This represents the frequency of deletions, which increases from left to right.
- Y-axis: Emotion Index (Numeric): Numeric values for Emotion Index, ranging from 0 to 4. Each number likely corresponds to an emotion category:0: Neutral, 1: Sad, 2: Angry, 3: Calm, 4: Happy.

The color bar on the right side maps the Emotion Index (Numeric) to specific colors are:

- Dark Purple: Represents Neutral (0).
- Blue: Represents Sad (1).
- Cyan: Represents Angry (2).
- Green: Represents Calm (3).
- Yellow: Represents Happy (4).

We can see that most of the points are concentrated at lower DelFreq values (0-20).
As DelFreq increases beyond 20, the frequency of emotions diminishes, with very few data points above 50 DelFreq.
From the neutral emotion, we can see that it appears across a wide range of DelFreq values, including high frequencies (e.g., around 120 and 140). This indicates that neutral emotions are more common at higher deletion frequencies.
The other emotions (Sad, Angry, Calm, and Happy) occur mostly at lower DelFreq values (0–40). Also, calm and Happy seem to dominate when DelFreq is very low.
This suggests that higher deletion frequencies may correlate with a neutral emotional state, while more expressive emotions occur at lower deletion frequencies.

Finally, plot 4 indicates that as the relative frequency of backspace key usage increases, the Emotion Index, which might represent the level of emotional engagement, decreases. This inverse relationship suggests that increased backspace key usage, potentially representing hesitation, uncertainty or error correction during typing, is associated with lower emotional engagement or arousal.